Nishanth Tatapudi
Bangalore, India
Open to relocate to Germany
Email: niktatapudi@gmail.com
Phone: +91 89516 56008
LinkedIn: Nishanth Tatapudi

Dear Hiring Manager,

I am writing to express my interest in the Software Test Engineer position in the automotive domain. With over 8 years of experience in automotive verification and validation, I bring strong expertise in HV & LV inverter systems testing for electric vehicles, working across OEM and non-OEM electric mobility platforms.

In my current role as Technical Leader – Systems at BorgWarner (Power Electronics Features & Solutions Testing), I validate HV inverter software for the Mercedes-Benz eATS platform (MMA & EVA2M). I analyze system requirements (ASPICE SYS.2/SYS.3) and develop test cases (SYS.4/SYS.5) using black-box techniques, then execute smoke, regression, and scenario-based testing on LV/HV benches and E-Dyne setups. I have validated key functions like inverter controls, PSOL (Position Sensor Offset Learning), flux-based and power-based torque estimation (FSL2), electrical machine offset learning, and rotor temperature monitoring(FSL2). Through structured regression planning, defect triage, and root-cause analysis with cross-functional teams, I helped reduce software defect leakage by <10% and improved release readiness and traceability using Polarion and JIRA.

Previously at SPANIDEA Systems, I validated the Ather Drive Controller (LV inverter) with a focus on traction performance, thermal stability, FMS protection features (throttle faults, OCP(HW/SW), motor stall, voltage/thermal protections), and UDS diagnostics. I executed performance and WOT test campaigns and verified Ride/Eco/Sport/Warp mode tuning across 50+ scenarios to strengthen controller robustness and reduce field issues. At Global Edge, I performed HIL-based VECU validation for Volvo Trucks across multi-ECU networks (J1939/J1587), validating features such as ACC, PTO, ACB, air suspension interlocks, and engine speed control - contributing to reduction in cross-ECU system defects.

I am ISTQB® certified and TÜV Rheinland certified in High Voltage Electrical Safety, and I bring a strong electrical/electronics foundation that helps me understand both hardware behavior and control software interactions in HV environments. I also have intermediate German proficiency and am committed to continuous improvement.

I would welcome the opportunity to discuss how my experience in inverter and ECU validation, ASPICE-compliant testing, and HIL/PHIL execution can support your e-mobility programs. Thank you for your time and consideration.

Yours sincerely,
Nishanth Tatapudi